

29. Design a 5G-Enabled IoT Framework for Campus Automation. (BL6)

Aim:

To design and simulate a 5G-enabled IoT framework for campus automation that connects smart devices and sensors through a high-speed 5G network for efficient monitoring, control and automation.

Objective:

- To connect campus sensors and IoT devices through a 5G network.
- To monitor campus parameters such as energy, lighting and occupancy.
- To provide fast and reliable communication between IoT devices and the control server.
- To automate campus facilities based on sensor information.
- To improve energy efficiency, safety and resource management.

Requirements:

- **Hardware Requirements:** Computer/Laptop; Minimum 4 GB RAM; IoT sensors; Microcontroller/IoT development board; 5G communication module or simulated 5G interface.
- **Software Requirements:** MATLAB / MATLAB Online; MATLAB Communication Toolbox (optional); Operating system: Windows/Linux.
- **Programming Requirements:** MATLAB; IoT data generation; 5G communication workflow; Automation and decision-making algorithm.

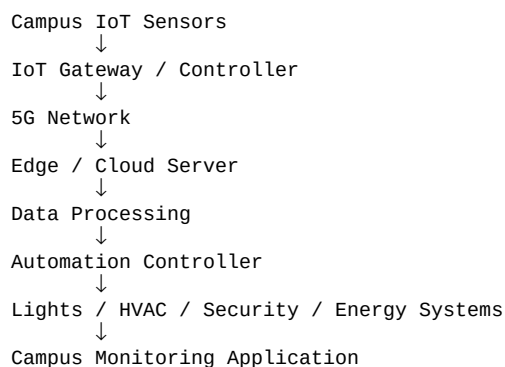
System Concept:

The 5G-enabled campus automation framework connects different IoT devices such as occupancy, temperature, lighting and energy sensors to a central control platform through a 5G communication network. The collected information is processed and appropriate control actions are generated automatically.

Basic Operation:

Sense Data → 5G Connectivity → Data Processing → Decision Making → Control Action → Monitoring

Simple Workflow:



Algorithm:

- Initialize the campus IoT system.
- Define the number of IoT devices and available resources.
- Generate or acquire sensor data from campus areas.
- Process and normalize the collected data.
- Establish 5G connectivity for IoT devices.
- Transmit sensor data to the edge/cloud server.
- Analyze the received data.
- Compare sensor values with predefined thresholds.
- Generate appropriate automation control commands.
- Send control commands to campus devices.
- Monitor the system performance continuously.
- Repeat the process for continuous campus automation.

MATLAB Code:

```

clc;
clear;
close all;

% Number of campus IoT devices
N = 10;

% Generate sample campus sensor data
temperature = 22 + 8*rand(1,N);
occupancy = randi([0 100],1,N);
energy = 50 + 100*rand(1,N);

% Threshold values
temp_threshold = 28;
occupancy_threshold = 60;
energy_threshold = 120;

% Automation decisions
AC = temperature > temp_threshold;
lights = occupancy > occupancy_threshold;
energy_alert = energy > energy_threshold;

% Display sensor data
disp('Campus IoT Sensor Data:');
disp('Temperature:');
disp(temperature);

disp('Occupancy:');
disp(occupancy);

disp('Energy Consumption:');
disp(energy);

% Display automation status
disp('AC Status (1 = ON):');
disp(AC);

disp('Lights Status (1 = ON):');
disp(lights);

disp('Energy Alert (1 = HIGH):');
disp(energy_alert);

% Plot temperature
figure;
bar(1:N,temperature);
xlabel('Campus Zone');
ylabel('Temperature (°C)');
title('Campus Temperature Monitoring');
grid on;

% Plot occupancy

```

```

figure;
bar(1:N,occupancy);
xlabel('Campus Zone');
ylabel('Occupancy (%)');
title('Campus Occupancy Monitoring');
grid on;

% Calculate automation efficiency
automated_zones = sum(AC | lights);
efficiency = (automated_zones/N)*100;

fprintf('Automation Activity = %.2f %%\n',efficiency);

```

Expected Output:

- Campus Temperature Monitoring Graph – shows temperature variations in different campus zones.
- Campus Occupancy Monitoring Graph – shows occupancy levels in different zones.
- AC and Lighting Status – indicates whether devices should be switched ON or OFF based on sensor conditions.
- Energy Alert – identifies zones with high energy consumption.
- Automation Activity – indicates the percentage of zones requiring an automated action.

Sample Result:

Campus IoT Sensor Data:

```

Temperature: 24.5  27.2  29.1  25.6  30.0 ...
Occupancy:   45    72    85    30    65 ...
Energy:      82    95   130    70   145 ...

```

```

AC Status (1 = ON):
0  0  1  0  1 ...

```

```

Lights Status (1 = ON):
0  1  1  0  1 ...

```

```

Energy Alert (1 = HIGH):
0  0  1  0  1 ...

```

```
Automation Activity = 60.00 %
```

Note: Actual values may change because sensor values are randomly generated in the simulation.

Advantages:

- Provides fast and reliable IoT connectivity using 5G.
- Supports real-time campus monitoring.
- Enables automatic control of campus facilities.
- Improves energy efficiency and resource utilization.
- Supports a large number of connected IoT devices.
- Provides scalable communication for smart campus applications.

Applications:

- Smart classroom automation
- Smart lighting and HVAC control
- Campus energy monitoring
- Smart security and surveillance
- Smart parking management

- Environmental monitoring
- Smart laboratory automation

Result:

Thus, a 5G-enabled IoT framework for campus automation was designed and simulated successfully. The framework collects campus sensor information, communicates it through 5G connectivity and automatically controls campus facilities according to the monitored conditions.